Plant Soil (2020) 448:231–251
https://doi.org/10.1007/s11104-019-04416-z

REGULAR ARTICLE

Metamodelling a 3D architectural root-system model to provide a simple model based on key processes and species functional groups

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Received: 5 September 2019 / Accepted: 26 December 2019 / Published online: 21 January 2020
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Abstract
Aims The architecture of root systems determines where and how much resources plants can extract from the soil, how they compete for soil resources, and how they interact with soil organisms. We aimed to develop a 3D root-system model called RSCone for future inclusion into multispecies individual-based canopy models suitable to design integrated weed management or intercropping strategies.
Methods We (1) proposed a conceptual root-system model consisting of empirical equations predicting root-system envelope, root length and biomass distribution from environmental conditions, species and plant stage, (2) calibrated the model from simulations with an existing architectural root-system model (ArchiSimple) to benefit from its knowledge on root functioning and the many parameterized crop and weed species, (3) identified the root-system architectural processes driving the key root state variables and (4) established species functional groups with Principal Component Analyses and clustering.
Results RSCone consists of 17 equations and 14 parameters; it was calibrated for 22 weeds and 22 crop species and varieties. Six species functional groups were established, depending on their family (*Poaceae*, *Fabaceae*, other), root-extension rates, specific root length (SRL) and the time to reach maximum SRL.
Conclusion RSCone is ready to be included into multispecies (crop and/or weed) dynamics models.

Keywords Calibration · Weed · Cover crop · ArchiSimple · Simulation

Responsible Editor: W Richard Whalley.

Electronic supplementary material The online version of this article (https://doi.org/10.1007/s11104-019-04416-z) contains supplementary material, which is available to authorized users.

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Introduction

Understanding root system development and growth, as well as root distribution in the soil is crucial to understand plant functions and responses to environmental conditions. The architecture of the root system determines where and how much resources the plant can extract from the soil (Barber 1984), and thus how plants compete for soil resources such as nitrogen or water (Gérard et al. 2017). The root system also determines over which soil volume plants spread root exudates, and thus how plants interact with organisms, whether beneficial (e.g. microorganisms described in Moreau et al. 2019) or harmful (e.g. parasitic plants, Grenz et al. 2005).

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Many models have been developed to represent root development and growth and/or root distribution that we can classify into two main categories: density models and architecture models. Density models (see examples in Dupuy et al. 2010) aim to describe distributions of particular root attributes (usually length or biomass) in the soil, particularly across soil layers. They are particularly useful when the root system must be described at a large scale, i.e. the whole crop canopy. Most of them use simple mathematical formalisms, either spatial functions or differential equations when the dynamics are considered. Such models are usually used in crop models simulating homogenous monospecies canopies from soil and weather inputs (e.g. STICS, Brisson et al. 2003; APSIM, Keating et al. 2003). However, these models remain abstract, relatively far from the root structures and processes, and they are often impossible to up or downscale. It can be difficult to estimate their parameters, which often have no direct biological meaning and vary with numerous environmental factors as well as with plant phenology. Such models can thus not be used in heterogeneous, particularly multispecies canopies.

Architectural models (e.g., Pagès et al. 2014) explicitly represent root systems, usually in 3D, in which the individual roots are split into segments and have precise locations in space. Most of them are dynamic and developmental models and represent individual plants. Their basic concepts are rather concrete and simple, close to the biological processes, and fit nicely with above-ground function-structure plant models (see examples in Gaudio et al. 2019). But, the combination of several developmental sub-processes makes these models more complex than the density models. Moreover, the number of individual objects in a root system is usually very high in a mature plant, so that these models are restricted to individual plants to limit simulation time and computer memory. This point is a serious limitation when the objective is to investigate plant populations or communities with plant-plant interactions. This is particularly the case for multispecies crop mixtures or crop-weed canopies whose heterogeneity in terms of plant emergence date, location and morphology requires a 3D individual-based modelling approach to correctly simulate plant-plant interactions (Colbach et al. 2019; Gaudio et al. 2019; Munier-Jolain et al. 2013; Renton and Chauhan 2017).

New models, somewhere between these two categories, are required, as pointed by Pedersen et al. (2010). These could feed homogeneous-canopy models such as STICS or APSIM to improve the modelling of soil-plant interactions (mainly, uptake and use of soil resources) as well as 3D individual-based models which do not need to represent each leaf or root as function-structure models but specifically represent plant-plant interactions (mainly, the competition for soil resources among neighbour plants) (see examples in Gaudio et al. 2019). The new root models should be conceptually simple and also easy and quick to compute, particularly when aiming to include them in individual-based models.

This new root-system model could be either built de novo, starting from scratch, or by simplifying an existing complex and mechanistic root-architecture model. The latter allows recycling data collection, formalisation and evaluation. One simplification technique is metamodeling, which aims at emulating the original model, linking inputs and outputs by less detailed but faster equations fitted to a huge simulated data set. This principle has been successfully applied to various fields, e.g. root depth density distribution (Pagès et al. 2012), forest growth (Marie and Simioni 2014), bio-geo-chemical cycles (Villa-Vialaneix et al. 2012) or 3D individual-based light interception (Colas 2018). These empirical metamodels replace the initial mechanistic models, using the same inputs and producing the same outputs. The metamodeling procedure includes a sensitivity analysis to identify minor inputs that can be ignored and thus produces new knowledge on the key input variable driving the analysed system. However, this approach requires a huge simulation plan to cover every contingency, and the prediction quality may decline when inputs are close to the limits of the simulated input ranges (Colas 2018).

The aim of the present paper was to metamodel an existing 3D architectural root-system model (ArchiSimple, Pagès et al. 2014) in order to develop a simpler, faster 3D root-system model (called RSCone), suitable for future connection with functions of soil-resource uptake and use in population or individual-based models. The new model must consider the impact of environmental conditions and carbon resource supply on the root system, and predict the key state variables relevant for plant-soil and plant-plant interactions, i.e. the envelope of the root system as well as the distribution of root biomass and root length inside this envelope (Fig. 1.A).

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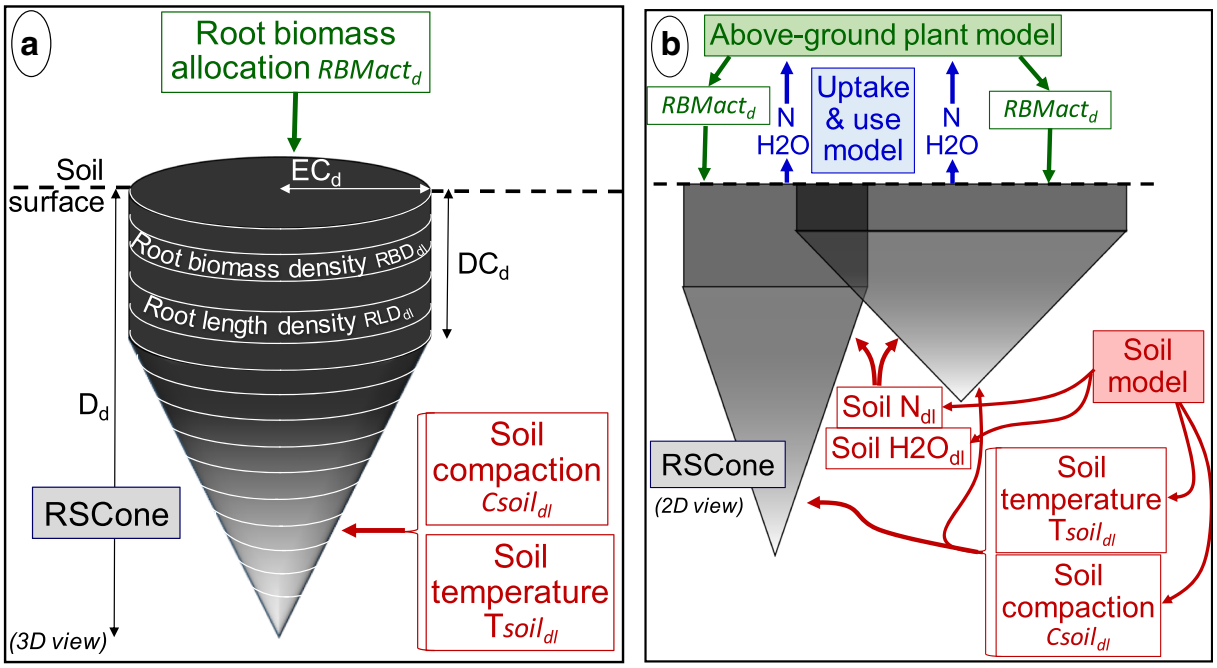

Fig. 1 Schematic representation of the RSCone root-system model as developed in the present paper (A) and how it can be integrated into an individual-based 3D canopy model in the future, simulating competition for soil-resource uptake in overlapping root-system zones (B), showing input variables, state variables per day d and soil layer l and connected models. Details on input and state variables are in Table 1 and Table 5, respectively (Nathalie Colbach © 2019)

The development of RSCone consists of several steps. First, we proposed its general structure based on expertise, literature and simulations with ArchiSimple to identify the root-system architectural processes driving the key state variables. RSCone was then calibrated based on simulations with ArchiSimple for a large range of contrasting species and species functional groups were established. Connecting RSCone with soil-resource uptake and above-ground functions will be done in future papers (Fig. 1.B).

The architectural model chosen as a starting point had to be sufficiently generic and realistic to be used in virtual experiments for contrasting crop and weed species, which is the case of ArchiSimple (Pagès et al. 2014). This model has been parameterized and tested for many contrasting species (including crops and weeds, monocot and dicots, legumes and non-legumes) and crop genotypes with different growth strategies (Bui et al. 2015; Drouet et al. 2005; Faverjon et al. 2019; Moreau et al. 2017a; Pagès et al. 2014; Pagès and Kervella 2018; Pagès and Picon-Cochard 2014).

Material and methods

The ArchiSimple model used for the virtual experiments

The ArchiSimple mode was presented in detail by Pagès et al. (2014). It is a dynamic architectural model, in which the root system is represented as a set of root segments (of some millimetres) and meristems (examples in Fig. 2). Each day, the virtual root system is modified through the combination of several developmental and growth submodels which formalize the main processes: emission of new adventitious roots from the shoot, elongation of existing roots, acropetal branching, radial growth and self-pruning following root decay.

Source and sink concepts are integrated in a very simple way. Each day, the resource supply (i.e. the root biomass provided by photosynthesis) is given as input. The demand for root biomass of all roots is then summed throughout the root system and compared to the available supply to determine a satisfaction coefficient as the ratio of the root biomass demand to the allocated biomass. Actual growth (i.e. elongation,

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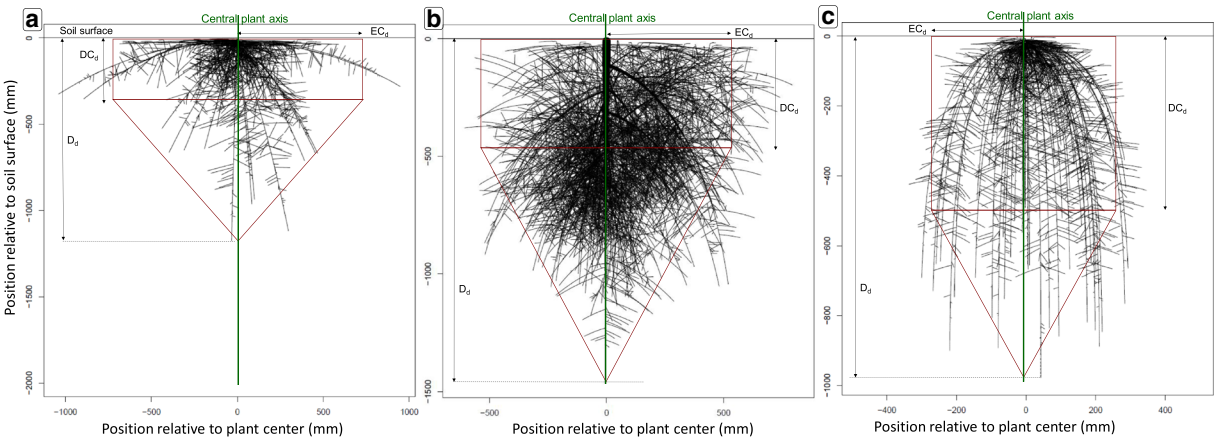

Fig. 2 Examples of root systems (A. Slow lateral roots and fast adventitious roots; B. Vigorous lateral roots, vigorous and early adventitious roots; C. Lateral roots with strong gravitropism) simulated with ArchiSimple and the corresponding RSCone envelope. In ArchiSimple, each root is a succession of segments whose advent and elongation are driven by processes such as emission of new adventitious roots from the shoot, elongation of existing roots, acropetal branching, radial growth and self-pruning following root decay. The RSCone envelope consists of a cylinder (depth DC_d, radius EC_d) on top a spilled cone (height = $D_d - DC_d$). Please note the difference in scales between the three graphs, a scaled representation can be found in section D.3 in supplementary material online. (Loïc Pagès & Nathalie Colbach © 2019)

branching etc) is calculated from potential growth using this coefficient, as well as soil temperature, mechanical constraints (Table 1) and species parameters (Table 2). These parameters are species properties implicated in the processes driving root architecture and their number is rather small (22) but the detailed representation of the root-system architecture, locating and representing each individual root segment slows down simulations, with 20 s for a single plant over only 80 days on a 2 GHz processor with 16 Go RAM. Together with the need for memory for storing the various root segments data, this is incompatible with the thousands of plants that must be simulated over decades in a 3D individual based crop-weed model.

Proposing the conceptual simplified root-system model RSCone

The new simplified root-system model was called RSCone (Root System seen as a Cone). It is a numerical model simulating the root system of a plant at a daily time step, to make it compatible with modelling processes such as photosynthesis or soil-resource take-up, whether in homogeneous-canopy models such as STICS (Brisson et al. 2003) or in 3D individual-based multispecies models (Colbach et al. 2019). RSCone's root system is enclosed by an extending 3D envelope, resulting from the superposition of a cylinder sitting on top of a spilled cone (Fig. 1.A). This envelope represents the root influence zone of the plant (Casper et al. 2003), and the combination of the two volumes allows to cover the multitude of possible root-system shapes presented by the usual crop and weed species from agroecosystems (Kutschera and Lichtenegger 1960). Three state variables determine the envelope, i.e. the depth and extent (radius) of the cylinder-shaped part of the root system, and the total root-system depth (Fig. 1.A, Fig. 2).

Each day, the root-system envelope grows, depending on root-biomass allocation by photosynthesis, plant age, soil compaction and soil temperature as well as species parameters. Two key root attributes are specified within the envelope: root-biomass density (i.e. dry root biomass per unit soil volume inside the root-system

Table 1 Input variables illustrating environmental constraints on day d and in soil layer l (cm from soil surface) driving root-system growth and development in both ArchiSimple and RSCone

Variable	Meaning	Unit
$Csoil_{dl}$	Soil constraints due to soil compaction and gravel content in soil layer l, ranging from 0 = allowing no root growth to 1 = no effect on root growth	cm cm^{-1}
$RBMact_d$	Root biomass provided by photosynthesis since seed germination	g plant^{-1}
$Tsoil_{dl}$	Soil temperature in soil layer l	°C

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Table 2 Species parameters driving root growth in the architectural model ArchiSimple (based on Pagès et al. 2014), with range of variation of values (Bui et al. 2015; Drouet et al. 2005; Moreau et al. 2017a; Moreau et al. 2017b; Pagès et al. 2014; Pagès and Kervella 2018; Pagès and Picon-Cochard 2014) and present paper for the crop and weed species calibrated later for RSCone

Parameter	Meaning	Unit	Range of variation in species calibrated for RSCone		
			mean	min	max
dicot	Clade (0: monocotyledon, 1: dicotyledon)		0.85	0.00	1.00
emissratesem	Emission rate of seminal roots	day^{-1}	0.98	0.50	1.00
propdiamsem	Ratio of the diameter of seminal roots to maximum root diameter		0.95	0.60	1.00
nbmaxsem	Maximum number of seminal roots		1.02	1.00	2.00
ageadvemis	Plant age when it starts to emit adventitious roots	day	4574	5	10,000
distmaxadv	Maximum distance of adventitious roots to the root base	mm	11.5	10.0	80.0
emissrateadv	Emission rate of adventitious roots	day^{-1}	0.20	0.00	1.68
propdiamadv	Ratio of the diameter of adventitious roots to maximum root diameter		0.90	0.90	1.00
nbmaxadv	Maximum number of adventitious roots		17	0	148
dmin	Root-tip diameter below which elongation is impossible (diameter of the finest roots)	mm	0.16	0.08	0.36
dmax	Maximum tip diameter of primary roots	mm	1.00	0.44	2.06
elongvsdiam	Slope of the potential elongation rate versus root-tip diameter	$mm \cdot mm^{-1} \cdot day^{-1}$	26.0	11.9	49.2
tgravi	Direction of tropism (plagiotropism; negative geotropism; positive geotropism; exotropism)		All positive geotropism		
igravi	Intensity of tropism		0.0045	0.0020	0.0080
durdp	Plant age when meristems are mature	days	5.00	5.00	5.00
ibd	Inter-primordium distance	mm	2.02	0.56	3.74
dldm	Average ratio of the diameter of the daughter root to that of the mother root	$mm\ mm^{-1}$	0.32	0.12	0.52
vard	Relative variation of the daughter root diameter		0.28	0.10	0.67
tmd	Root tissue density	$g \cdot cm^{-3}$	0.11	0.03	0.30
growthdur	Slope of the growth duration versus squared tip diameter	$day \cdot mm^{-2}$	323	170	350
lifeexp	Slope of the life duration versus tip diameter and tissue mass density	$day \cdot mm\ mg^{-1}$	2513	2500	2800
radgrowth	Proportionality coefficient between section area of the segment and the sum of distal section areas (radial growth coefficient)		0.13	0.00	0.25

envelope) and root-length density (i.e. the cumulated root length per unit soil volume) which is derived from biomass density via the specific root length (SRL, i.e. root length per unit of root biomass) and also depends on plant age. These attributes are essential to simulate extraction of soil resources (Aziz et al. 2017; Malagoli and Le Deunff 2014) and/or the attachment of root pathogens or parasites (Grenz et al. 2005), once RSCone will be included in homogeneous-canopy models or 3D individual-based models. This connection is not part of the present paper, which is a necessary preliminary step. When included in 3D individual-based models, RSCone will allow simulating below-ground plant-plant interactions by predicting where root systems of neighbouring plants overlap and compete for soil-resource uptake (Fig. 1.B). Above-ground plant-plant competition for light will affect root-biomass allocation resulting from photosynthesis.

Building RSCone

To determine the shape of the equations driving the extension of the root-system envelope, biomass and root length from plant age and environmental conditions, contrasting root systems were simulated with ArchiSimple. To ensure that the equations were sufficiently generic and applied to both crops and weeds, we chose root systems that contrasted the most for the functions to be analysed. The choice was based on the expert knowledge of the ArchiSimple authors and

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previous ArchiSimple simulations (e.g., Pagès and Kervella 2018; Pagès and Picon-Cochard 2014). Each root system was repeated on five plants to account for stochastic effects in ArchiSimple, and simulated over 80 days, in non-limiting conditions, i.e. at optimal soil temperature, without soil compaction or gravel content, and without root biomass provision being limited by photosynthesis. In terms of actual plant age, these 80 days under optimal temperature conditions (e.g., 24 °C for maize) cover the duration of root-system growth of most species. For each root-system state variable, we here presented the two or three most contrasting examples to demonstrate the adequacy and genericity of the equations to be used in RSCone. Simulated data were analysed, using lm() and nls() functions of R (R Core Team 2016) for linear and non-linear regressions, respectively.

Parameterizing RSCone

A series of 35 weed and crop species (including 10 genotypes for *Pisum sativum*) were parameterized for ArchiSimple (Table 2), from our own experiments and literature (Bui et al. 2015; Drouet et al. 2005; Moreau et al. 2017b; Pagès et al. 2014; Pagès and Kervella 2018; Pagès and Picon-Cochard 2014) (section B.1 online). RSCone parameters were estimated through calibration, by fitting the RSCone equations to virtual data simulated with ArchiSimple. For each species, three plants (considered as replicates) were simulated over 70 days, in non-limiting conditions in terms of soil structure, soil temperature and biomass provision. The simulations of section "Building RSCone" had shown that fewer replicates and days were sufficient to estimate species parameters. Then, the RSCone parameters were estimated by fitting the RSCone equations to the simulated output of each plant, using lm() and nls() of R (R Core Team 2016). These parameter values were then averaged over the three plants of each species. One RSCone parameter necessary to simulate the effect of soil constraints on root-system state variables could be directly calculated from an ArchiSimple parameter (Table 2.B). Two others, base and optimal temperatures, were mostly taken from crop and weed models (e.g., STICS, Brisson et al. 2003; Colbach et al. 2014; FlorSys, Colbach et al. 2019).

As ArchiSimple includes stochastic processes, the calibration script was run twice over 21 weed species and 14 crop species, among which 10 pea varieties (Table 6) and Pearson correlation coefficients were calculated between the parameter values of the two runs for each RSCone parameter, using the cor() function of R.

Functional species groups

Correlations among RSCone parameters were investigated with a Principal Component Analysis (PCA), followed by a Ward ascendant hierarchy classification to cluster crop and weed species into functional groups, depending on their root-system parameters, using the PCA() and hclust() functions of the FactoMineR package (Lê et al. 2008) of R (R Core Team 2016). To identify which ArchiSimple parameters were the major determinants of root-system characteristics in RSCone, the ArchiSimple parameters were projected onto the PCA axes. Pearson correlation coefficients were also calculated between ArchiSimple and RSCone parameters.

Results

Daily steps in RSCone

In addition to species parameters (Table 3), RSCone needs daily soil temperature and mechanical constraints as well as root biomass provision resulting from photosynthesis as inputs (Table 1), i.e. the same inputs as ArchiSimple. The equations are run daily to drive root-system growth for the virtual plant in RSCone and are presented in Table 4, the list of state variables describing the root system in Table 5. Time is in days since seed germination; the slowing down of root growth and development by too cold or too hot conditions is accounted for by temperature-dependant corrective functions.

The following sections explain the various equations and how they were chosen.

Depth of the root system envelope

The total depth of the root system is the distance from soil surface to the peak of the spilled cone (Fig. 2). In non-limiting conditions (i.e. no soil compaction, optimal soil temperature, non-limiting root biomass provision), the potential root-system depth $Dpot_d$ solely depends on plant age for a given species. To identify a generic equation linking root-system depth to plant age, we ran ArchiSimple simulations for a series of root-system types contrasting in terms of emergence timing, root elongation

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Table 3 Species parameters driving root-system growth and development in RSCone, with values for maize (*Zea mays*) as an example

Parameter	Meaning	Maize values	Unit	Estimation method[§]
A. Timing and extent of root system				
t0	Timing of root growth onset	7	Days since germination (under optimal temperature)	Durdp + (1/emissratesem)
t0cyl	Delay from **t0** to when the cylinder part of the root system starts to grow in depth	31	Days since **t0** (under optimal temperature)	Fit eq. 6, or min of d-**t0** where $E_d > 0.01$ cm
tmax	Timing of root growth end	infinite	Days since germination (under optimal temperature)	Fit eq. 4
tSRLmax	Timing of maximum specific root length (SRL)	59.2	Days since germination (under optimal temperature)	Fit eq. 16
Emax	Maximum extent (radius) of the root system	24.7	cm	Fit eq. 5
RBDmax	Root biomass density in cylinder part of the root system, disregarding constraints	$2.28 \cdot 10^{-4}$	$g \cdot cm^{-3}$	Calculated from eq. 13
rD	Speed at which root-system depth increases	1.11	$cm \cdot day^{-1}$ (under optimal temperature)	Fit eq. 4
rE	Speed at which root-system extent (radius) increases	0.608	$cm \cdot day^{-1}$ (under optimal temperature)	Fit eq. 5
rCD	Ratio of speed at which depth of cylinder-shaped part of root system increase vs speed of total root-system depth increase	0.144	$cm \cdot day^{-1}\ cm^{-1}\ day^{-1}$ (under optimal temperature)	Mean of DC_d/D_d if d > **t0** + **t0cyl**
srl0	Specific root length when roots start to grow	552	$cm \cdot g^{-1}$	Fit eq. 16
srlmax	Maximum specific root length	3170	$cm \cdot g^{-1}$	Fit eq. 16
B. Effect of environmental conditions				
Pmax	Ability of roots to penetrate the soil	1	cm/cm, ranging from 0 (unable) to 1 (no limitation)	If dmin x < 0.3, **Pmax** = 0.7 Else if dmin <1.2 **Pmax**= 0.6 + 0.33 dmin Else **Pmax** = 1
Tbase	Base temperature for growth and development	8	°C	Literature
Topt	Optimal temperature for growth and development	24	°C	Literature

[§] Directly calculated from ArchiSimple parameters (Table 2), estimated by fitting equations of Table 4 to state variables (Table 5) simulated with ArchiSimple, or calculated from state variables simulated with ArchiSimple

and emission rates, importance of gravitropism etc. Figure 3 shows the three most contrasting systems. The first system represents the linear growth of a tap-rooted system as can be found in pea (*Pisum sativum*). The taproot started growing after a lag of several days following germination, and continued to grow until the end of the simulation. The second system represents a determinate growth pattern (Shishkova et al. 2008). Elongation stopped to grow after approximately two months. In the third system which is common in *Poaceae* species, the seminal root was the deepest until day 40, and was then outgrown by faster growing adventitious roots, without stopping until the simulation end.

The same equation (eq. 4 of Table 4) satisfactorily fitted all three behaviours (Fig. 3). The lag from germination to root-system extension in depth (**t0**) could directly be linked to two ArchiSimple parameters, as the sum of the plant age when meristems are mature plus the inverse of the emission rate of seminal roots (Table 3).

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Table 4 Comprehensive list of equations for simulating the root system in RSCone, with a daily time step (d = days since germination) and according to soil layer l (cm from soil surface). State variables in standard black with definitions in Table 5, species parameters in **bold blue** with definitions in Table 3, inputs in *italic* with definitions in Table 1

N°	Process	Equation
Environmental constraints on root-system growth		
1	Root growth limitation due to soil temperature	If $Tsoil_{dl} < \mathbf{Tbase}$ then $rTsoil_{dl} = 0$ else if $Tsoil_{dl} < \mathbf{Topt}$ then $rTsoil_{dl} = \sin(0.5 \cdot \pi \cdot (Tsoil_{dl} - \mathbf{Tbase}) / (\mathbf{Topt} - \mathbf{Tbase}))$ else $rTsoil_{dl} = 1$ $rTsoil_d$ = mean of $rTsoil_{dl}$ with $l \leq D_{d-1}$
2	Root growth limitation due to soil structure	If $Csoil_{dl} \geq \mathbf{Pmax}$ then $rPen_{dl} = 0$ else $rPen_{dl} = 1 - Csoil_{dl} / \mathbf{Pmax}$ $rPen_d$ = mean of $rPen_{dl}$ with $l \leq D_{d-1}$
3	Root growth limitation due to soil constraints (temperature and compaction)	$rSoil_d = rTsoil_d \cdot rPen_d$
Potential root system dimensions (disregarding soil constraints)		
4	Potential root system depth	If $d < \mathbf{t0}$, then $Dpot_d = 0$ else if $d < \mathbf{tmax}$, then $Dpot_d = \mathbf{rD} \cdot (d - \mathbf{t0})$ else $Dpot_d = \mathbf{rD} \cdot (\mathbf{tmax} - \mathbf{d0})$
5	Potential root system lateral extent (radius of the cylinder-shaped part of root system)	if $d \leq \mathbf{t0}$, then $ECpot_d = 0$ else if $d < \mathbf{t0} + \mathbf{Emax} / \mathbf{rE}$, then $ECpot_d = \mathbf{rE} \cdot (d - \mathbf{t0})$ else $ECpot_d = \mathbf{Emax}$
6	Potential depth at which root system extent (radius) is maximal (depth of cylinder-shaped part of root system)	If $d < \mathbf{t0} + \mathbf{t0cyl}$, then $DCpot_d = 0$ else $DCpot_d = (d - \mathbf{t0} - \mathbf{t0cyl}) \cdot \mathbf{rD} \cdot \mathbf{rCD}$
Actual root system dimensions (accounting for soil constraints)		
7	Actual root system depth	$D_d = Dpot_d \cdot rSoil_d$
8	Actual depth at which root-system extent (radius) is maximal	$DC_d = DCpot_d \cdot rSoil_d$
9	Actual maximum root-system extent (radius)	$EC_d = ECpot_d \cdot rSoil_d$
10	Lateral root system extent (radius) in soil layer l	If $l < DC_d$, then $E_{dl} = EC_d$ else if $l < D_d$, then $E_{dl} = EC_d \cdot (1 - \frac{l + 0.5 - DC_d}{D_d - DC_d}$ else $E_{dl} = 0$
Root-system biomass		
11	Volume of root layer l inside the root-system envelope	If $l < DC_d$ then $vol_{dl} = \pi \cdot EC_d^2 \cdot l$ else if $l < D_d$ then $vol_{dl} = \pi \cdot EC_d^2 \cdot DC_d^2 + \pi \cdot 0.25 \cdot EC_d^2 \cdot (D_d - DC_d) - \pi \cdot 0.25 \cdot EC_d^2 \cdot (D_d - l)^3 / (D_d - DC_d)^2$ else $vol_{dl} = 0$
12	Potential biomass density per soil layer (disregarding insufficient photosynthesis but accounting for soil constraints)	If $l < DC_d$ then $RBDpot_{dl} = \mathbf{RBDmax}$ else if $l < D_d$ then $RBDpot_{dl} = \mathbf{RBDmax} \cdot (D_d - l) / (D_d - DC_d)$ else $RBDpot_{d.} = 0$
13	Potential root-system biomass (disregarding insufficient photosynthesis but accounting for soil constraints)	If $d < \mathbf{t0}$, then $RBMpot_d = 0$ else $RBMpot_d = \mathbf{RBDmax} \cdot 0.25 \cdot \pi \cdot E_d^2 \cdot (3 \cdot DC_d + D_d)$
14	Actual root biomass (accounting for soil and photosynthesis constraints)	$RBM_d = \min(RBMpot_d, RBMact_d)$ If $RBMpot_d > 0$ then $rPhoto_d = RBM_d / RBMpot_d$ else $rPhoto_d = 0$
15	Actual root biomass density in soil layer l (accounting for soil and photosynthesis constraints)	$RBD_{dl} = rPhoto_d \cdot RBDpot_{dl}$
Root length density (accounting for soil and photosynthesis constraints)		
16	Specific root length density	if $d \leq \mathbf{t0}$, then $srl_d = \mathbf{srl0}$ else if $d < \mathbf{tsrlmax}$, then $srl_d = (\mathbf{srlmax} - \mathbf{srl0}) \cdot (d - \mathbf{t0}) / (\mathbf{tsrlmax} - \mathbf{t0}) + \mathbf{srl0}$ else $srl_d = \mathbf{srlmax}$
17	Root length density in soil layer l	If $l \leq D_d$, then $RLD_{dl} = srl_d \cdot RBD_{dl}$ else $RLD_{dl} = 0$

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Table 5 Root-system state variables on day d predicted by RSCone for a virtual plant

Variable	Meaning	Drivers (other than species)	Unit
$Csoil_d$	Average soil constraints due to soil compaction and gravel content, ranging from 0 = allowing no root growth to 1 = no effect on root growth	Soil constraints in each soil layer (input)	cm cm^{-1}
DC_d	Actual depth of the cylinder-shaped part of the root system (considering soil constraints)	Potential depth at which the root system width starts to shrink, soil constraints	cm
$DCpot_d$	Potential depth of the cylinder-shaped part of the root system (disregarding soil constraints)	Plant age	cm
D	Actual root-system depth (considering soil constraints)	Potential root-system depth, soil constraints	cm
$Dpot_d$	Potential root-system depth (disregarding soil constraints)	Plant age	cm
EC_d	Actual extent (radius) of the cylinder-shaped part of the root system (considering soil constraints)	Potential root-system extent, soil constraints	cm
$ECpot_d$	Potential extent (radius) of the cylinder-shaped part of the root system (disregarding soil constraints)	Plant age	cm
E_{dl}	Root-system extent (radius) in soil layer l	Actual root-system extent, soil layer	cm
RBD_{dl}	Root biomass density in soil layer l	Actual root biomass in soil layer l, volume of layer l inside root-system envelope	g·cm^{-3}
RBM_{dl}	Actual root biomass in soil layer l	Actual root biomass	g plant^{-1}
RBM_d	Actual root biomass	Potential root biomass, root biomass allocation by photosynthesis (input)	g plant^{-1}
$RBMpot_d$	Potential root-system biomass	Actual root-system dimensions, plant age	g plant^{-1}
RLD_{dl}	Root length density in soil layer l	Root biomass density in soil layer, specific root length	cm·cm^{-3}
$rPen_{dl}$	Relative reduction in root-system expansion due to root penetration resistance exerted by the soil (due to soil compaction and gravel content) in soil layer l	Penetration resistance of soil layer l (input)	cm cm^{-1}
$rPen_d$	Average relative reduction in root-system expansion due to root penetration resistance exerted by the soil (due to soil compaction and gravel content)	Relative reduction in root-system expansion due to root penetration resistance in soil layer l	cm cm^{-1}
$rPhoto_d$	Relative reduction in root biomass due to insufficient photosynthesis	Potential root biomass, root biomass provided by photosynthesis	g g^{-1}
$rSoil_d$	Relative reduction in root-system expansion due to soil constraints (temperature, compaction)	Relative reduction in root-system expansion due to mechanical constraints, idem due to temperature	cm cm^{-1}
$rTsoil_{dl}$	Relative reduction in in root-system expansion due to low soil temperature in soil layer l	Soil temperature (input)	cm cm^{-1}
$rTsoil_d$	Average relative reduction in in root-system expansion due to low soil temperature	Relative reduction in in root-system expansion due to low soil temperature in soil layer l	cm cm^{-1}
srl_d	Specific root length	Plant age	cm·g^{-1}
vol_{dl}	Volume of soil layer l inside root-system envelope	Root-system dimensions	cm^3

Lateral root-system extension

Lateral extent defines the radius of both the cylinder and the cone base (Fig. 2). It was calculated to enclose 95% of the root segments and tested on a series of contrasting root systems, three of which can be seen in Fig. 2. In the first case (Fig. 2.A), the lateral roots emerging from the radicle extended slowly and were overtaken by adventitious roots with a higher elongation rate after 25 days. The second case (Fig. 2.B) presented a rather constant expansion rate due to vigorous lateral roots as well as vigorous and precocious adventitious roots. The third root system (Fig. 2.C) presented a significant gravitropism in lateral roots. The curvature of these

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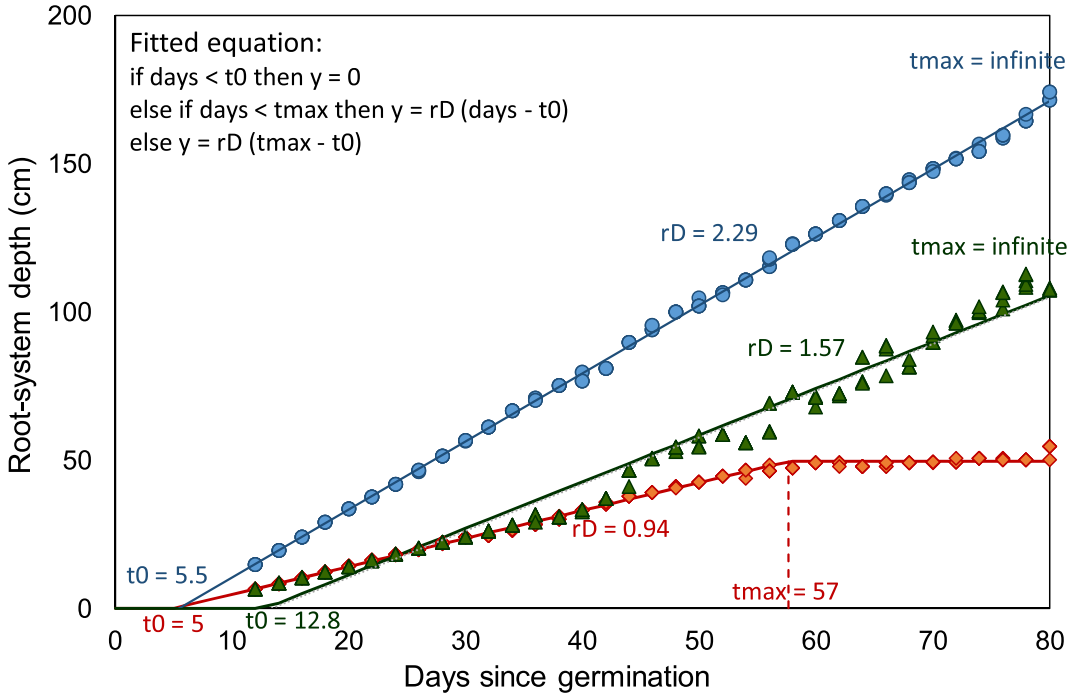

Fig. 3 Root depth versus plant age simulated with ArchiSimple for three contrasting root-system types (● tap-rooted system, ♦ determinate growth pattern, ▲ determinate seminal followed by indeterminate adventitious growth) in non-limiting conditions (no soil compaction, optimal soil temperature, non-limiting root biomass provision), with five replicates for each system. The pictures of the root systems can be found in section D.1 online (Loïc Pagès & Nathalie Colbach © 2019)

lateral roots induced a gradual decrease of the lateral extension rate with time. As for root-system depth, a broken-stick linear regression was sufficiently generic to approximate all situations (eq. 5 in Table 4, section C.2 online).

Depth of the cylinder-shaped part of the root system

The same principle was used to determine the shape of the equation driving the depth of the cylinder-shaped part of the root system (Fig. 2) from plant age and species parameters. ArchiSimple was run for a series of contrasting root systems. The depth of the cylinder was defined by first locating % root segments that were located furthest from the central plant axis, and then calculating the 95-percentile of their depths (Fig. 2). After a lag of a few days, this depth increased linearly with plant age, irrespective of the tested root system. The first root system (Fig. 2.A) presented vigorous adventitious roots emitted from day 12 onwards, with a low gravitropism common in *Solanaceae* species. In the second system (Fig. 2.B), lateral root-system expansion was driven by the vigorous acropetal lateral roots of the taproot common among *Brassicaceae* such as *Brassica napus*. In the third system common in *Poaceae* species (Fig. 2.C), the lateral expansion started almost immediately, with vigorous adventitious roots and a strong gravitropism.

The analysis of the ratio of the cylinder depth vs the root-system depth showed that this ratio was constant regardless of plant age (section C.2 online). The lag between seed germination and the time when the cylinder started to grow in depth (**t0+ tcyl0)** was longer than the lag for total root-system depth increase (**t0**). Consequently, RSCone uses the same extension rate (**rD**) for both cylinder and total depths, multiplying the result by a parameter (**rCD**), and increases the initial lag by a further lag (**tcyl0**) for cylinder depth growth (eq. 6 in Table 4).

Effect of soil constraints

The growth of root-system depth is usually limited by two soil characteristics: temperature and soil resistance to root penetration, due to soil compaction. These characteristics are given as input variables, with one value per 1-cm thick soil layer (Table 1). They are transformed into relative coefficients, reflecting relative reduction in

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root-system expansion, ranging from 0 (i.e. no root growth) to 1 (no limitation in root growth due to soil compaction, too cold or to hot temperature).

The effect of temperature is supposed to follow a sinusoidal curve (section C.1 online), increasing from 0 when soil temperature ($Tsoil_{dl}$) equates the base temperature (**Tbase**) of the species to 1 when the optimal species temperature (**Topt**) is reached (eq. 1 in Table 4). This effect is calculated for each 1-cm thick soil layer and then averaged over the current root depth.

The effect of the mechanical constraint (eq. 2 in Table 4) was inspired by literature (Bengough and Mullins 1990) and is assumed to be a linear function of the soil compaction ($Csoil_{dl}$), decreasing from 1 when soil offers no resistance to roots, to 0 when soil structure allows the species maximum penetration ability (**Pmax**). This empirical parameter **Pmax** assesses the species ability to penetrate compacted soils and was calculated from a ArchiSimple parameter (i.e. maximum root tip diameter dmax): **Pmax** increases from 0.7 (root elongation stops when soil compaction reach 70%) to 1 cm/cm (root elongation stops when the soil is totally compacted) when the maximum root tip diameter increases from 0.3 to 1.2 mm (section C.1 online). As for temperature, the soil resistance effect is calculated for each soil layer and then averaged over root depth.

The two effects are combined (eq. 3 in Table 4) and then applied to potential root depth to get actual root depth (eq. 7). The same principle is applied to the depth and radius of the cylinder-shaped part of the root system (eq. 8 and 9). Once the actual dimensions of the root envelope are known, the root-system lateral extent (i.e. radius) inside the cone can be calculated as a linear regression (eq. 10), making the extent shrink from its maximum value EC_d at the top of the cone (soil layer $l = DC_d$) to zero at the base of the cone, i.e. depth of the root system (layer $l = D_d$).

Root biomass density

RSCone does not consider the horizontal variations of biomass distribution, only its vertical variation along soil depth. For this purpose, we made a large number of simulations of various types and ages of root systems, assuming that biomass provision to roots resulting from photosynthesis was non-limiting, and visually explored the variations of biomass density on graphs (root biomass per 1-cm soil layer divided by soil layer volume inside root-system envelope, eq. 11 in Table 4). This exploration showed that the biomass distribution could greatly vary, depending on the root system architecture. The simplest generic approach was to consider that root biomass density was constant in the cylinder-shaped part of the root system and then linearly decreased to zero at the tip of the cone (eq. 12). This equation only needed a single parameter, i.e. maximum root biomass density. If a greater precision of root biomass vs depth is needed, more complicated equations can be used but these need one or two additional parameters (e.g. a two-parameter Weibull equation, section C.4 online).

When investigating root biomass density variations with plant age in contrasting root systems, we could not find a relationship with plant age. So, **RBDmax** simply is the mean of the maximum root biomass densities estimated by fitting eq. 12 to plants older than 25 days, to eliminate stochasticity in values which is very high in young plants.

Root-system biomass

The potential root biomass $RBMpot_d$ depends on root-system dimensions, integrating root biomass density over the root-system volume (eq. 13 in Table 4). The actual root biomass of the day also depends on biomass provision resulting from photosynthesis which is an input variable in both ArchiSimple and RSCone (Table 1). Actual biomass is the minimum between the potential and the actual provided biomass (eq. 14 in Table 4). The ratio of these two biomasses ($rPhoto_d$) is then used to determine the day's actual root biomass density, accounting for insufficient photosynthesis (eq. 14 in Table 4). Insufficient biomass provision to roots from photosynthesis does not affect root-system dimensions. Indeed, insufficient resources primarily limit the emission and elongation of fine roots rather than those of primary root axes, which define the shape and size of the root-system envelope.

Specific root length

In order to calculate root length density from biomass density, we also need the specific root length (SRL) over time, i.e. the total root length divided by the total root biomass. As previously, we tested different root systems, and two contrasting examples can be found in Fig. 4. Both root systems showed the same basic pattern in the change in SRL. In the root system with thick roots and radial growth (such as *Pisum sativum*), SRL started

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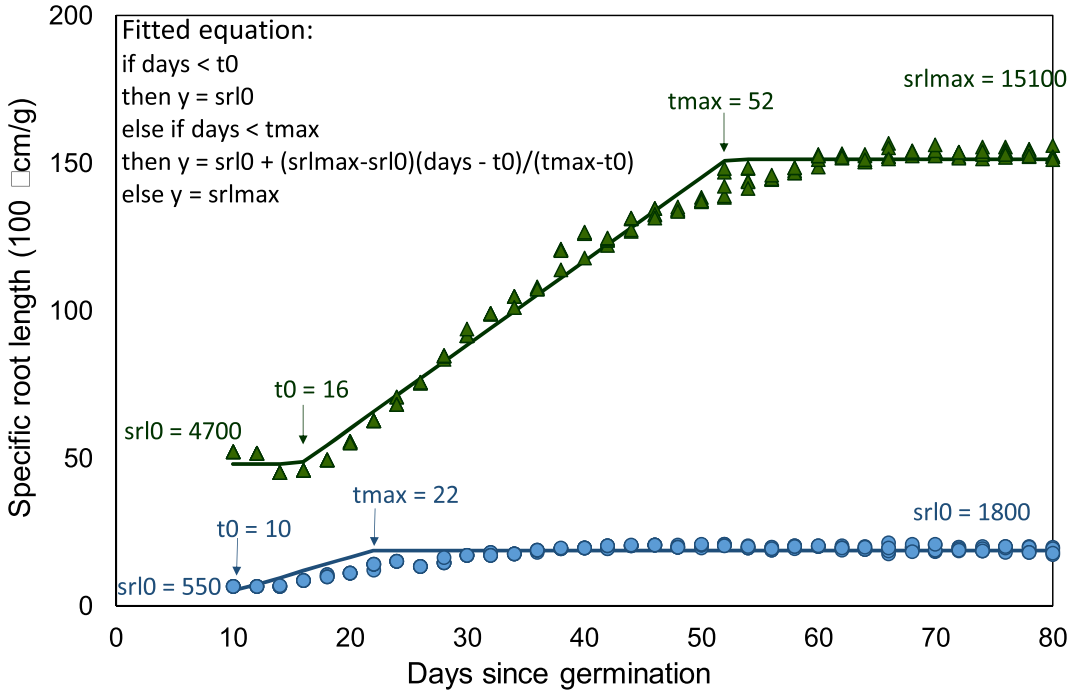

Fig. 4 Specific root length (cumulated root length divided by root biomass) versus plant age simulated with ArchiSimple for two contrasting root-system types (● thick root with radial growth, ▲ thin roots without radial growth) in non-limiting conditions (no soil compaction, optimal soil temperature, non-limiting root biomass provision), with five replicates for each system. The pictures of the root systems can be found in section C.4 online (Loïc Pagès & Nathalie Colbach © 2019)

at a low level, increased during 10 days until it reached a constant value approximating three times the initial value. The second system, with thinner roots and without radial growth (e.g. *Poaceae*), started with nearly ten times the SRL of the first root system. The increase lasted for more than a month, before reaching a level again approximately three times the initial value.

Whatever the root-system type, the change in SRL with plant age can be modelled with a broken-stick regression (eq. 16 in Table 4), depending on initial and final values (**srl0**, **srlmax**) and the timing of the onset and end of the period of SRL increase (**t0**, **tsrlmax**). Root length density in a given soil layer l can then simply be calculated by multiplying root biomass density on the layer by the day's specific root length (eq. 17).

Species typology

Thirty-five species of 11 families, among which 22 weed and 12 crop species, including 10 genotypes of *Pisum sativum* (Table 6), were calibrated for RSCone by fitting the equations of Table 4 to root-system architecture simulated with ArchiSimple. The comparison of the parameter values of the two calibration runs showed them to be very stable (Pearson correlation coefficients ranging from 0.96 to 1 between the parameter values of the two runs, data not shown), despite ArchiSimple stochasticity. With the exception of the timing of root-system growth (**t0**), the RSCone parameters varied considerably among the calibrated species (Fig. 5).

Some RSCone parameters were correlated. Generally, the species and varieties with the greatest lateral root-system extent (**Emax**) were also those with the fastest growth speed, in terms of extent (**rE**), depth (**rD**) and cylinder-depth (**rCD**) (Fig. 6.A). They also started the growth of the cylinder-shaped part of the root system earlier (opposite to **t0cyl**). Species and varieties with a large specific root length SRL during early growth stages (**srl0**) were also able to reach much larger SRL later in their life (**srlmax**), but tended to be less resistant to soil compaction (low **Pmax**). There were no other strong correlations among RSCone parameters (Fig. 6.C). Particularly, the onset of root-system growth

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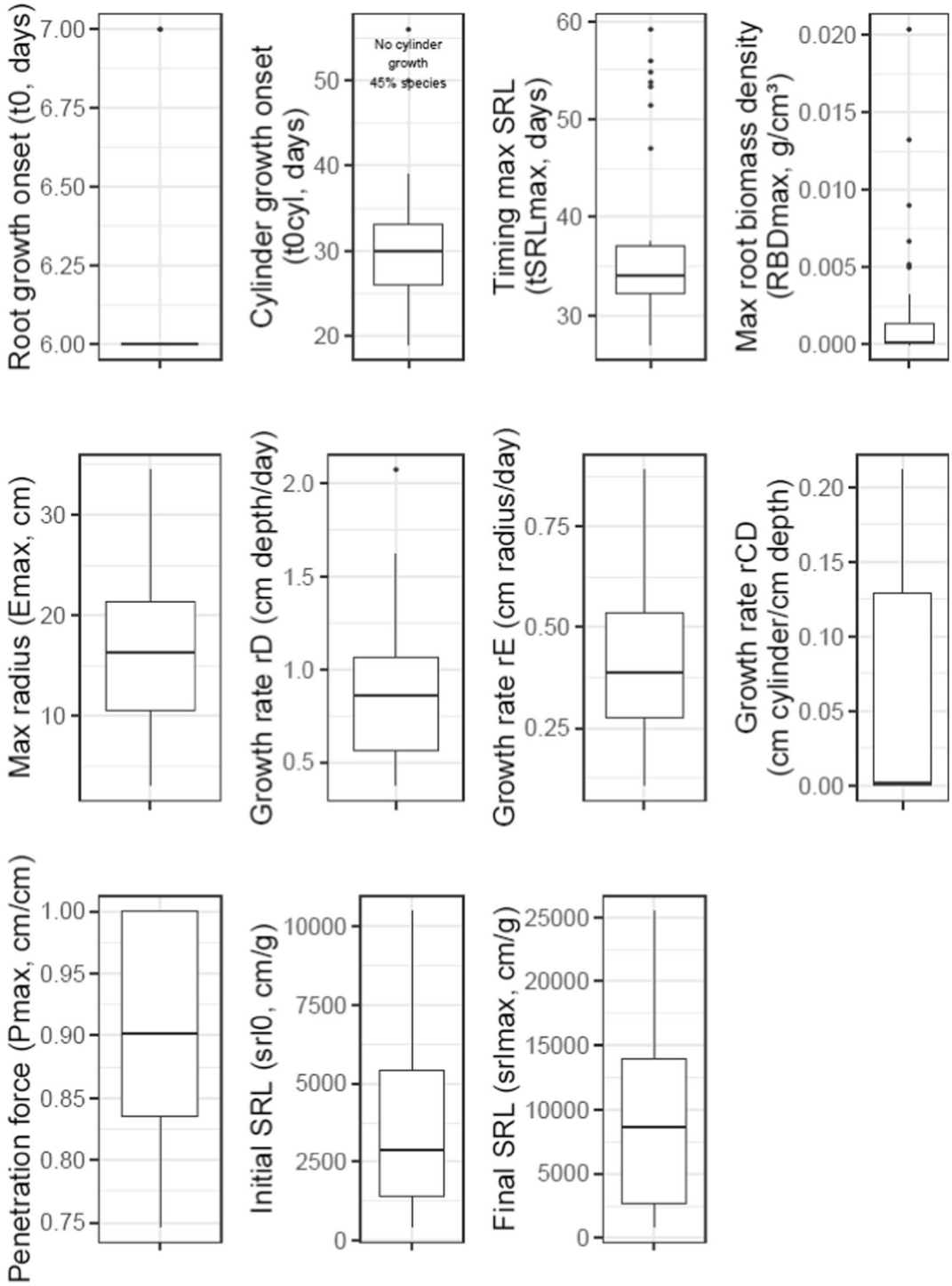

Fig. 5 Distribution of species parameters driving root-system state variables in RSCone. Parameters were estimated by fitting the equations of Table 4 to simulations with ArchiSimple parameterized for the 45 species and varieties of Table 6. For further information on RSCone parameters, see Table 3

(**t0**) and the age when the maximum SRL was reached (**tSRLmax**) and the maximum root biomass density (**RBDmax**) were little correlated to any other parameter (Fig. 6.A and C).

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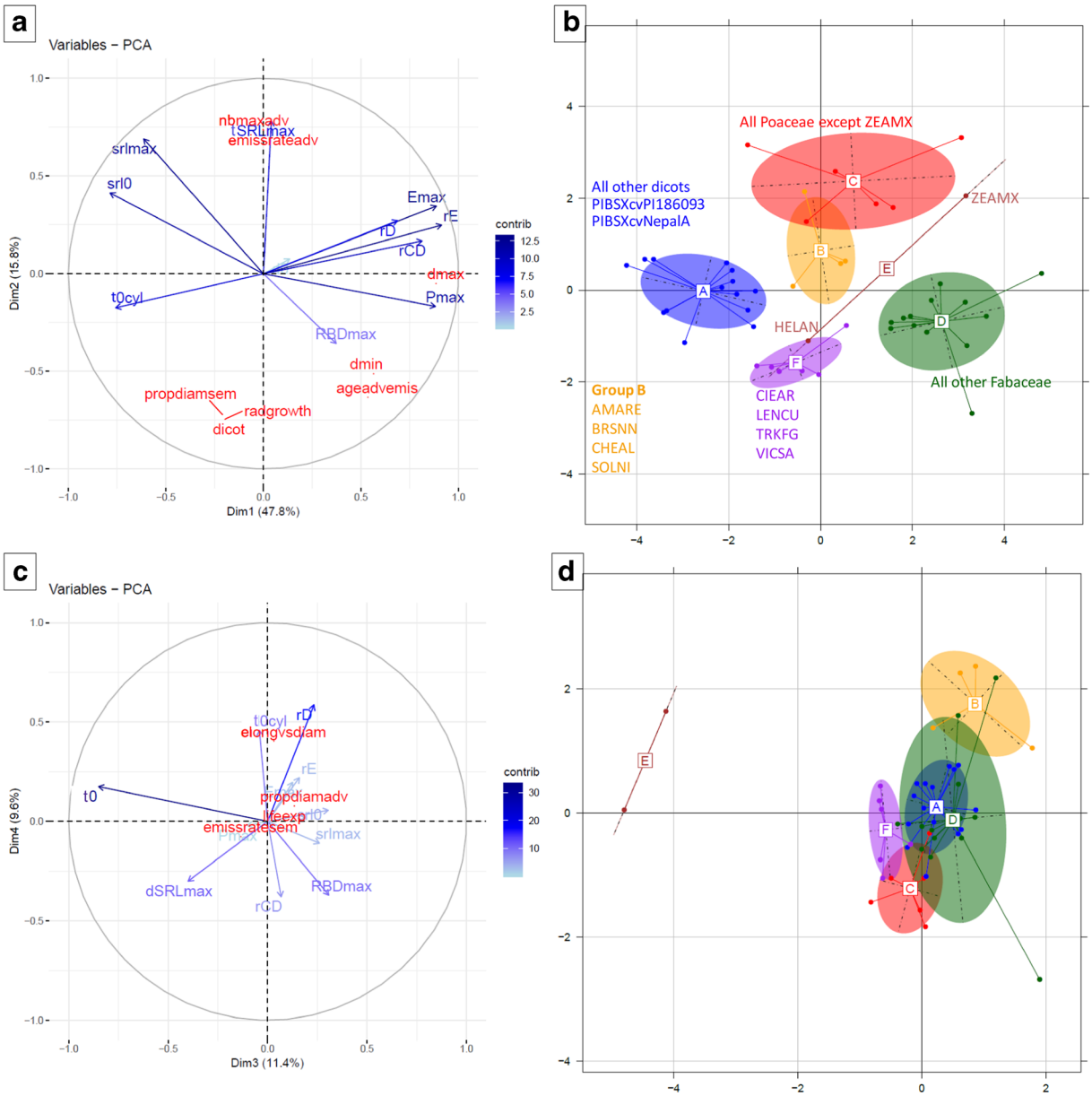

Fig. 6 Typology of crop and weed species based on a Principal Component Analysis on the RSCone parameters and the 45 species and varieties of Table 6. Correlation circles with RSCone parameters in blue (see Table 3 for meaning), with the five most important ArchiSimple parameters projected in red for the first two (A) and the last two axes (C), colouring variables according to their contribution to the analysis. Crop and weed species individuals clustered into groups, following a Ward ascendant hierarchy classification for the first two (B) and the last two axes (D). Species names are EPPO codes (Nathalie Colbach and Loïc Pagès © 2019)

Six functional groups in terms of root-system growth could be determined from the RSCone parameters (Fig. 6.B). Group C consisted of all tested *Poaceae* species except maize (ZEAMX) and these were essentially characterised by a delayed maximum SRL (**tSRLmax**), and a large and fast lateral root-system extent (**Emax, rE**). Non-legume dicotyledonous species were essentially split into two groups. Group B was closest to *Poaceae* species, but reaching maximum SRL slightly earlier (**tSRLmax**) and a fast vertical rather than lateral

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root-system extent (**rD** on Fig. 6.D). It consisted of only four species, from four different families. All the other non-legume dicots (except sunflower, HELAN) were included in group A, characterized by slow root-system growth (low **rE, rD, rCD**), low resistance to soil compaction (low **Pmax**) but higher specific root length (**srl0**).

Fabaceae (legume species) were also essentially split into two groups. Group F comprised four (low **rE, rD**) legume species with a slow root-system growth. The other legumes were all in group D which was roughly the opposite of group A in terms of attributes. The only exception were the two pea varieties with a very low seed mass (Table 6), i.e. PI186093 and NepalA, which were in group A with the majority of non-legume dicotyledonous species.

Group E, consisting of the tall summer annuals *Helianthus annuus* (HELAN) and *Zea mays* (ZEAMX), was particular insofar as it shared many characteristics with the *Poaceae* group C and the *Fabaceae* group F (Fig. 6.B) but it differed from all groups with its slightly later onset of root-system growth (**t0**, Fig. 6.D).

Which root-architecture features drive root-system extension?

When using ArchiSimple parameters to discriminate the species, the species typology was much simpler (section B.2 online), with roughly Poaceae, Fabaceae and non-legume dicotyledons. *H. annuus* and *Z. mays* were again set apart, differing from all other groups and from each other.

Among the 22 ArchiSimple parameters, only a few were highly correlated to RScone parameters (Fig. 6), and thus need to be estimated with greatest care in the future. The identified correlations were consistent with the rules underlying root-system growth in ArchiSimple. Species with larger maximum diameter primary-root tips (dmax) had roots that penetrated better into compacted soil in RSCone (**Pmax**), and their root system grew faster and larger (**Emax, rD, rE**, **rCD**) though their specific root length SRL tended to be lower (**srl0, srlmax**) (Fig. 6.A). Low SRL in RSCone was associated to large minimum diameters in primary roots tips in ArchiSimple (large dmin) and an early onset of adventitious-root emission (low ageadvemis). Species that, in RSCone, reached their maximum SRL late (**tsrlmax**) were usually monocots (opposite to dicot), presented a large maximum number of seminal roots (nbmaxsem), with a high emission rate of adventitious roots (emissrateadv), a low radial growth coefficient (opposite to radgrowth) and narrow seminal roots (opposite to propdiamsem).

The other parameters were less closely linked (Fig. 6.C). Species that, in RSCone, grew faster in terms of root-system depth (**rD**) also presented a high root elongation rate relatively to root-tip diameter in ArchiSimple (elongvsdiam). Other links could not be seen on the dimension presented in Fig. 6. For instance, a late onset of root growth in RSCone (**t0**) was linked to a low emission rate of seminal roots in ArchiSimple (emissratesem, Pearson correlation coefficient = 1).

Discussion

A fast and simple dynamic model, with a low number of parameters

This modelling approach is a new means to make a simple and dynamic representation of the root system of individual plants. Such representations are necessary to take into account several issues related to soil-plant relations, especially when considering multispecies and heterogeneous canopies. While 3D-individual based models all consider plant-plant competition for light, competition for soil resources is still rarely considered (Gaudio et al. 2019), and the present model will be an essential step toward this goal, once it is connected to above-ground plant models and functions for soil-resource uptake and use.

Simplicity is an important criterion to achieve this task, for several reasons. In most cases, it is important to use a model that is easy to understand, from a conceptual point of view, and easy and quick to program and to compute. Moreover, the number of parameters and state variables describing the root system must be as low as possible. RSCone only uses 14 parameters compared to ArchiSimple's 22 (or several dozens of parameters as in some models, Dunbabin et al. 2013), and these parameters all have a straightforward biological meaning. Most of them can be calibrated independently from real data (measured in greenhouse or field experiments) or, even better, from virtual data produced

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from models such as ArchiSimple as we did here. The parameters can thus be estimated from different data sources, which is a major advantage compared to conventional metamodels whose structure and parameter values are very dependent on the source model used to build the metamodel. The main simplification though lies in the simplification of the root-system structure, simulating an envelope filled in by root biomass and length, instead of explicitly simulating the size and location of each root segment as in ArchiSimple (Pagès et al. 2014) or other models (Dunbabin et al. 2013). Despite the simplification, the RSCone structure allows to predict the same key state variables for plant-plant interaction while dividing the simulation time by more than 20 compared to ArchiSimple. Moreover, the daily steps are sufficiently independent to allow increasing complexity, e.g. a more precise albeit more parameter-hungry equation can be used to simulate root biomass density with depth.

Several authors have emphasized this need for simple models (e.g., Pedersen et al. 2010) and rather simple models of the root system have been developed during the last years (Bonneu et al. 2012; de Willigen et al. 2002; Dupuy et al. 2010) based on differential equations to describe the dynamics of root density. But, these model do not discriminate the root zone of individual plants, they are calibrated from root density maps observed in pure stands in fields, and are thus not adapted to simulate plant-plant interactions in heterogeneous multispecies canopies (see section "Integration into homogeneous-canopy models and 3D individual based models" below). RSCone is substantially different, and closer to the model of Pedersen et al. (2010), but with several key advantages (see next section).

However, as for root density models (see examples in Dupuy et al. 2010), the RSCone representation does not make sense for the very young root systems before they have reached a significant extension and/or branching. This limit has though little effect in many conditions. Indeed, nitrogen availability has little effect on very young plants (e.g., before 150–200 °Cdays approximately, Fayaud et al. 2014) and young roots are still too small to attract many parasites (Gibot-Leclerc et al. 2012). Conversely, lack of water can limit pre-emergent growth (Bouaziz and Bruckler 1989), and radicle growth must thus be modelled (Dürr et al. 2001; Gardarin et al. 2012).

Another structural limit of RSCone for 3D canopy modelling is its assumption that the plant's root system is symmetric, with homogeneous root characteristics per horizontal soil layer. Both soil structure and resources not only vary vertically but also horizontally (Metcalfe et al. 2019; Roger-Estrade et al. 2004) and root systems tend to grow toward resources or away from constraints (Cassab et al. 2013). If the aim is to work at such a detailed scale (rather than at the cropping-system scale for instance), then function-structure models might be more appropriate (see examples in Gaudio et al. 2019).

Simplify virtual rather than actual reality, or why a metamodel rather than an empirical model?

Pedersen et al. (2010) built their model from scratch, using field observations. While this has the advantage of investigating reality, the model's structure and domain of validity are highly conditional on the studied species and environmental conditions. Using a detailed architectural root-system model based on generic and universally valid processes as virtual reality results in a model with a larger domain of validity and whose structure should be sufficiently generic to cover most plant species. Pagès et al. (2012) used a similar approach, building root density functions from intensive simulation and sensitivity analysis of a very simple root-system architecture model. Here, we went a step further, starting with a more complex root-system architecture model, testing a larger diversity of root system types, predicting a larger number of root-system state variables and accounting for the effect of environmental limiting factors.

Basing the simple root-system model on a process-based mechanistic model allowed checking which limiting factors should be considered in the simplified progeny model. For instance, the rules driving root-system architecture in non-legumes were shown to be little affected by nitrogen nutrition status (Brun et al. 2010; Moreau et al. 2017a). The root-system parameters in RSCone thus do not need to vary with nitrogen availability even though the root-system state variables will be affected indirectly via root biomass provision (Brun et al. 2010; Moreau et al. 2017a) which is an input of RSCone. So, RSCone is probably able to account for differences in plant root distribution in soil due to nitrogen limitation resulting from competition for nitrogen.

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By analogy, it may also allow reflecting differences in plant root distribution due to competition for light (the most frequent limiting resource in crop-weed canopies). Indeed, shading decreases both plant biomass production and biomass provision to roots (Brouwer 1962).

Starting from the mechanistic model also improves the quality of the parameter measurements. It is easier to measure, for instance, minimum root-tip diameter in controlled conditions than the empirical parameters of RSCone (e.g., root penetration force) or Pedersen's model (2010). Moreover, basing the new model on an existing model allows to capitalize knowledge. Indeed, ArchiSimple has been parameterized for a wide range of species and genotypes (Dicot vs. monocot, legume vs. non legume, weed vs crop…) (Bui et al. 2015; Drouet et al. 2005; Moreau et al. 2017b; Pagès et al. 2014; Pagès and Kervella 2018; Pagès and Picon-Cochard 2014) and still continues to be parameterized for new species and genotypes (Faverjon et al. 2019) and be linked to new models (Gérard et al. 2017). All these species and genotypes can therefore be easily, rapidly and cheaply parameterized for RSCone via simulations (running approximately for an hour per species).

Evaluating RSCone's prediction ability

The main drawback of the present modelling approach is its conditionality on the prediction quality of the underlying mechanistic model, here ArchiSimple. As ArchiSimple only considers root-system architecture, disregarding its functions as well as other plant organs, it is very difficult to evaluate with independent field data. But, field trials have shown that ArchiSimple well predicts specific root length, root-system depth and root biomass at two dates (Pagès and Picon-Cochard 2014), or that key species parameters such as root-tip diameter measured in 60 species were similar in two sites contrasting in terms of region, habitat, soil, and weather (Pagès and Kervella 2018).

Transforming ArchiSimple into the simpler RSCone which can be more easily connected to homogeneous-canopy models and 3D individual based models opens the door for checking the consistency of its predictions. Indeed, these models predict realistic crop and weed state variables which are comparable to usual field measurements (e.g. plant biomass, crop yield) and will allow an indirect evaluation of RSCone. Such a comparison does not actually check whether the details of the root system are correctly predicted but rather whether the root-system variables relevant for the plant functioning and reproduction are. And, to think even further, once these canopy-scaled models are used to evaluate and design cropping systems, it will be essential to verify whether they (as the result of many interacting submodels) allow to take the right decisions in terms of crop management. This is part of an ongoing work and the first results show that including RSCone into the crop-weed dynamics model FlorSys (Colbach et al. 2014; Colbach et al. 2019) improves the prediction of crop and weed species densities, biomasses and yields, even without considering competition for soil resources (Pointurier 2019).

Integration into homogeneous-canopy models and 3D individual based models

Though RSCone does not explicitly model plant-plant interactions or the effects of pedoclimate and cropping techniques, its inputs and outputs are tailored to make this possible when linked to other models (Fig. 1.B). Once connected to soil models predicting soil temperature and structure from pedoclimate and cropping techniques, RSCone allows simulating the effect of the latter two environmental variables on root systems. Once the actual biomass provision (i.e. $RBMact_d$ in eq. 14 in Table 4) is predicted by ecophysiological functions considering light interception and shading by neighbouring plants, the effect of plant-plant competition for light on the root systems will be included. Finally, the predicted root-system state variables will allow to determine where and how much soil resources are extracted from the soil, and thus how roots of neighbouring plants interact.

When connecting RSCone to models simulating homogeneous monospecies canopies (e.g. STICS, Brisson et al. 2003; APSIM, Keating et al. 2003), RSCone simply represents an average plant and the horizontal root-system extension can be ignored. However, the vertical distribution of root length density and root depth are key state variables to simulate the extraction of soil resources and tolerance to drought (e.g. Barber 1984). But, root density models (see examples in Dupuy et al. 2010) would allow reaching the same goal.

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The main target of RSCone are 3D individual-based models simulating heterogeneous multispecies canopies from pedoclimate and cropping techniques, particularly those running over several years (Colbach et al. 2019; Gaudio et al. 2019; Renton and Chauhan 2017). Indeed, in the past, the investigation of plant-plant competition focused on light as soil resources were usually sufficient in the intensively managed cropping system of temperate climates. But, recently because of environmental and health issues, national and European policies require farmers to switch to integrated crop production using fewer or no chemical inputs. In this context, once connected to above-ground plant models and functions for soil-resource uptake and use, RSCone will allow to tackle plant-plant competition for soil resources, by representing the reduction of resource uptake of plants whose root systems overlap in resource-limited soil areas (Fig. 1.B). Including this process in crop models is essential to evaluate innovative cropping systems based on biological pest regulation in virtual experiments, e.g. when simulating crop-weed canopies to identify stratgies for controlling weeds which are the most harmful pest for crop production (Oerke 2006) or when investigating intercrops as one of the key levers of agroecological crop production (Hauggaard-Nielsen et al. 2008).

Similar architectural rules can lead to different root systems

The RSCone model allowed to identify some of the emergent properties of the underlying root-system architectural model. We were able to identify functional species types in terms of root-system attributes and to link those to key ArchiSimple parameters, i.e. close to the biological processes driving root-system growth and architecture. Interestingly, the RSCone parameters (which result from the aggregation of several processes) allowed to discriminate sub-groups among non-legume dicots, among legumes and even among pea genotypes that could not be identified with the process-close ArchiSimple parameters. It is though difficult to compare our RSCone or ArchiSimple functional groups to those established on root-system observations from fields. Indeed, the latter are the result of both root-architecture rules and biomass provision whereas the former solely account for architecture rules, either directly (ArchiSimple) or indirectly (RSCone). But, the correlations between RSCone and ArchiSimple parameters demonstrated architecture rules observed in situ and/or included in ArchiSimple were preserved in RSCone such as the decrease in specific root length with increasing root diameter (Ma et al. 2018) (https://www.nature.com/articles/nature25783#supplementary-information), the difference between *Poaceae*, legumes and non-legumes in terms of root system morphology or, more interestingly, similar discriminations inside these groups, e.g. *Vicia sativa* (VICSA) behaving differently from other legumes (Bodner et al. 2013). Moreover, our analysis allowed us to identify the key ArchiSimple parameters driving root-system architecture, showing for instance that the minimum root-tip diameter Dmin is more influential than the maximum root-tip diameter Dmax, or that radial growth (radgrowth) substantially modifies the dynamics of specific root length SRL.

A good knowledge of the inter- or intra-specific differences in root traits allows hypothesizing differences among species or genotypes in their ability to take up nutrients, depending on their location and mobility. For instance, species or genotypes with a high root ability to penetrate the soil and a high rate of root-system increase in depth (right hand of Fig. 6.B) are assumed to better access the deepest soil resources, such as water and leached nitrate (Dunbabin et al. 2013). Besides, species or genotypes with a high specific root length (top left hand of Fig. 6.B) are generally associated to a high uptake efficiency of relatively immobile nutrients, such as phosphorus (Eissenstat 1992; Pagès 2011), due to a high colonization of the soil volume increasing the probability to reach poorly mobile nutrients. However, only considering the values of the parameters is insufficient to definitely conclude on the outcome of plant-plant competition. As other factors interact (such as the date of emergence of plants, their phenology, their access to light, the effects of techniques and pedoclimate), simulations are essential to better understand and predict interactions among plants.

Acknowledgements This project was supported by INRA and the French project CoSAC (ANR-15-CE18-0007). They authors are grateful to Maé Guinet, Marlène Lefèbvre, Mathieu Chanis and Juliette Senèze for contributing to the provision of datasets on legume species and genotypes.

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Appendix

Table 6 List of crop and weed species used in ArchiSimple simulations to estimate RSCone parameters

Species	EPPO code	Variety	Family	Status	Seed mass (mg)
Alopecurus myosuroides	ALOMY		Poaceae	Weed	2.3
Amaranthus retroflexus	AMARE		Amaranthaceae	Weed	0.38
Avena fatua	AVEFA		Poaceae	Weed	18.5
Brassica napus	BRSNN	Kadore	Brassicaceae	Crop	4.4
Bromus hordeaceus (mollis)	BROMO		Poaceae	Weed	2.77
Capsella bursa-pastoris	CAPBP		Brassicaceae	Weed	0.14
Centaurea cyanus	CENCY		Asteraceae	Weed	5.3
Chenopodium album	CHEAL		Amaranthaceae	Weed	0.56
Cicer arietinum	CIEAR	Twist	Fabaceae	Weed	382
Echinochloa crus-galli	ECHCG		Poaceae	Weed	2.24
Fallopia convolvulus	POLCO		Polygonaceae	Weed	6.52
Galium aparine	GALAP		Rubiaceae	Weed	7.37
Geranium molle	GERMO		Geraniaceae	Weed	1.07
Glycine max	GLXMA	Sultana	Fabaceae	Crop	216
Helianthus annuus	HELAN		Asteraceae	Crop	41.0
Lens culinaris	LENCU	Anicia	Fabaceae	Crop	31.0
Lupin album	LUPAL	Feodora	Fabaceae	Crop	218
Phaseolus vulgaris	PHSVX	Flavert	Fabaceae	Crop	162
Pisum sativum	PIBSX	Amino	Fabaceae	Crop	203
Pisum sativum	PIBSX	Austin	Fabaceae	Crop	292
Pisum sativum	PIBSX	Cameor	Fabaceae	Crop	152
Pisum sativum	PIBSX	Cuzco 1	Fabaceae	Crop	182
Pisum sativum	PIBSX	Isard	Fabaceae	Crop	199
Pisum sativum	PIBSX	Kayanne 1[§]	Fabaceae	Crop	na
Pisum sativum	PIBSX	Kayanne 2	Fabaceae	Crop	216
Pisum sativum	PIBSX	Kayanne 3	Fabaceae	Crop	249
Pisum sativum	PIBSX	L1073	Fabaceae	Crop	253
Pisum sativum	PIBSX	Livioletta	Fabaceae	Crop	125
Pisum sativum	PIBSX	NepalA	Fabaceae	Crop	41.7
Pisum sativum	PIBSX	PI186093	Fabaceae	Crop	116
Polygonum aviculare	POLAV		Polygonaceae	Weed	1.52
Polygonum persicaria	POLPE		Polygonaceae	Weed	1.9
Senecio vulgaris	SENVU		Asteraceae	Weed	6.52
Solanum nigrum	SOLNI		Solanaceae	Weed	1.90
Sonchus asper	SONAS		Asteraceae	Weed	0.26
Stellaria media	STEME		Caryophyllaceae	Weed	0.80
Teucrium botrys	TEUBO		Lamiaceae	Weed	0.30
Trigonella foenum-graecum	TRKFG	Fenufix	Fabaceae	Crop	0.40
Tripleurospermum inodorum	MATIN		Asteraceae	Weed	0.27
Triticum aestivum	TRZAX	Caphorn	Poaceae	Crop	42.0
Veronica hederifolia	VERHE		Plantaginaceae	Weed	3.52
Vicia faba	VICFX	Espresso	Fabaceae	Crop	359
Vicia narbonensis	VICNA	Clara	Fabaceae	Crop	231
Vicia sativa	VICSA	Candy	Fabaceae	Crop	82.0
Vulpia myuros	VLPMY		Poaceae	Weed	0.83
Zea mays	ZEAMX		Poaceae	Crop	252

[§] *P. sativum* cv Kayanne was grown three times, in three different experimental series

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